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Chapter 1
Probabilistic Guarantees

Abstract Here we show probabilistic guarantees for the success of the reasoning mechanisms on Computation-Activation Networks.

Topics: Chapter II.10

1.1 Fluctuations

1.1.1 Widths of the noise vector

1.1.2 Recovery guarantees given width bounds

1.2 Face recovery

Faces to the mean parameter represent the hard constraints in a model. Recovering the right face therefore corresponds with a correct identification of the hard constraints.

Lemma 1.1 *When generating data by a distribution $\mathbb{P}^*\left[X_{[d]}\right]$ with mean μ^* we always have*

$$\mu_D\left[L\right] \in \mathcal{F}(\mu^*)\,.$$

Proof. We always have $\mathbb{I}_{\neq 0}\left(\mathbb{P}^*\left[X_{[d]}\right]\right) \prec \mathbb{I}_{\neq 0}\left(\mathbb{P}^D\left[X_{[d]}\right]\right)$ when drawing data from $\mathbb{P}^*\left[X_{[d]}\right]$. Since the distribution $\mathbb{P}^*\left[X_{[d]}\right]$ has its mean on the face $\mathcal{F}(\mu^*)$, the image of its support is included in $\mathcal{N}_{\mathcal{F}(\mu)}$. Also the image of the support of $\mathbb{P}^D\left[X_{[d]}\right]$ is included therein and thus $\mu_D\left[L\right] \in \mathcal{F}(\mu^*)$. $\square$

Estimating the maximum entropy distribution with the data mean, we always estimate the face parameter as $\mathcal{F}(\mu_D)$. Since $\mu_D\left[L\right] \in \mathcal{F}(\mu^*)$ we have $\mathcal{F}(\mu_D) \subset \mathcal{F}(\mu^*)$. The constraints are therefore never underestimated.

Example 1.1 (Face recovery of bernoulli)

The mean parameter is parameter of Bernoulli distribution, the distribution of μ_D is the binomial distribution

$$\mu_D \sim \frac{1}{m} B(m, \mu^*)\,.$$

Face recovery:

- When $\mu^* \in \{0, 1\}$, then we always have face recovery and perfect recovery.
- When $\mu^* \in (0, 1)$, then we have not always face recovery. This case is investigated in the following.

Expectation of squared norm of the error: $\frac{1}{m}\mu^* \cdot (1 - \mu^*)$ We have for any $\delta > 0$ the chebyshev bound

$$\mathbb{P}\left[|\mu_D - \mu^*| > \delta\right] = \mathbb{P}\left[|\mu_D - \mu^*| > \delta^2\right] \leq \frac{\mu^* \cdot (1 - \mu^*)}{m\delta^2} \, .$$

Thus, choosing $\delta = \mu^*$, we have for any η with probability $1 - \eta$ face recovery provided that

$$m \geq \frac{1}{\eta} \cdot \frac{1 - \mu^*}{\mu^*} \, .$$

To show that the bound is tight we compute the exact probability of face recovery. The face recovery event is the complement of all datapoints being 0 or 1. The probability is therefore

$$1 - (1 - \mu^*)^m - (\mu^*)^m \, .$$

If μ^* is very small, then this is approximately $m \cdot \mu^* + O(\mu^{*2})$. Therefore we need $m \sim \frac{1}{\mu^*}$ samples for face recovery.